Benchmarking Spatio-Temporal Graph Neural Networks for Time Series Forecasting Across Heterogeneous Graph Domains

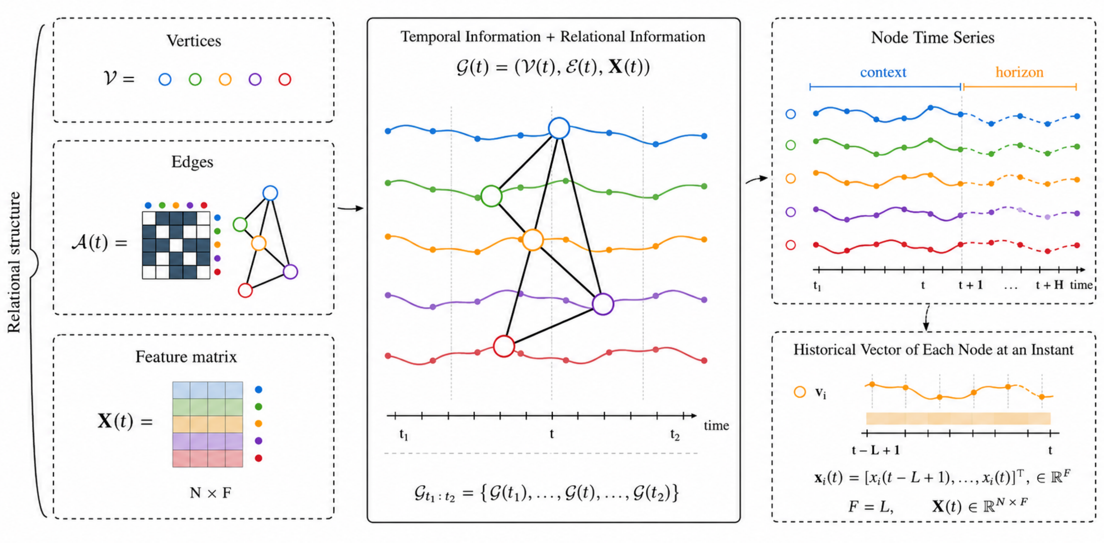
Gabriel F. Costa[1], Vander L. S. Freitas[1]

[1]Postgraduate Program in Computer Science, Federal University of Ouro Preto, Brazil

1 Introduction

1.1 Research Aim

- Graph time series arise in epidemiology, mobility, transportation and information systems, where node signals evolve over connected entities.
- STGNNs combine a spatial graph operator g_ϕ with a temporal module h_ψ, but published comparisons often mix datasets, splits, preprocessing, horizons and hyperparameters.
- This benchmark isolates architectural effects through a unified, controlled and reproducible protocol.



1.2 Contributions

How do convolutional, recurrent and attention-based STGNNs behave under the same protocol across heterogeneous graph domains and forecasting regimes?

1.3 GitHub Repository

Public implementation and experiments
github.com/gabrielxcosta/Spatiotemporal-GNN-Forecasting
Scan the QR code to access the repository, source code, and experimental material.

2 Literature Review

2.1 Graph Theory

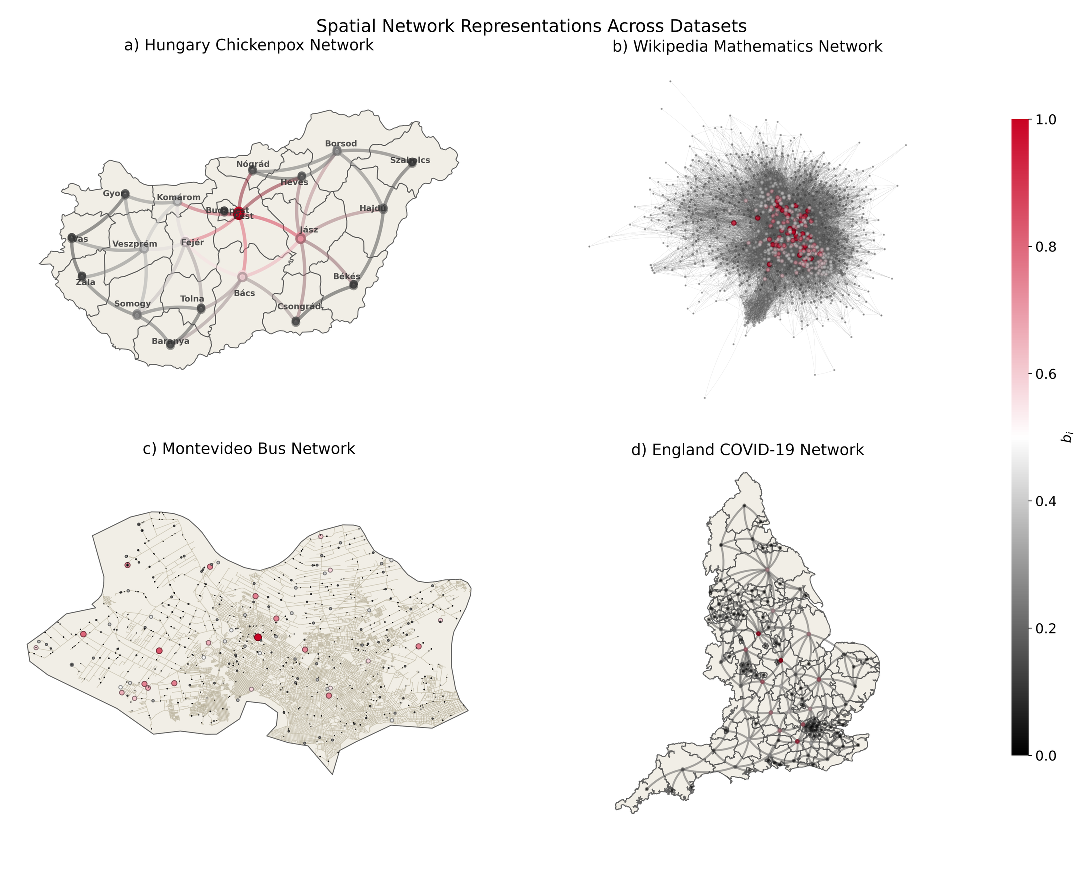
- Graph snapshot: $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), \mathbf{A}(t))$, with $|\mathcal{V}| = N$ fixed across time.
- Node signal tensor: $\mathbf{X}(t) \in \mathbb{R}^{N\times F}$; mini-batch input: $\mathbf{X} \in \mathbb{R}^{B\times L\times N\times F}$.
- Normalized Laplacian: $\mathcal{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$.
- Graph Fourier coefficients: $\hat{\mathbf{x}} = \mathbf{U}^\top\mathbf{x}$; spectral energy is grouped into low, mid and high bands $(\mathcal{E}_L, \mathcal{E}_M, \mathcal{E}_H)$.

2.3.1 Spatio-Temporal GNNs and Their Taxonomy

Family	Temporal module h_ψ	Main behavior
Attention	self-attention, temporal attention, transformer	Adaptive weighting of nodes/time steps; non-local dependencies.
Convolutional	temporal, gated, dilated or inception convolution	Parallel local filtering; controlled receptive field.
Recurrent	RNN/GRU/LSTM hidden-state updates	Sequential memory with graph propagation.

3 Materials & Methods

3.1 Datasets & Graph Analysis



Dataset	$\mathcal{G}_{type}$	D	w	Δt	T	$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{E}\rvert$	$\langle k\rangle$	ρ_{max}	$\mathcal{E}_L$	$\mathcal{E}_M$	$\mathcal{E}_H$	$\mathcal{E}_D \pm \sigma$
Chickenpox	S	N	N	W	517	20	41	4.10	6.00	0.4387	0.2785	0.2828	16.26 ± 17.35
WikiMaths	S	Y	Y	D	723	1,068	27,079	50.71	4.00	0.6562	0.1702	0.1736	629.89 ± 348.82
England	D	Y	Y	D	53	129	663.89	10.29	5.79	0.6370	0.1856	0.1774	70.33 ± 24.16
Montevideo	S	Y	Y	H	740	675	690	2.04	155.00	0.4267	0.2902	0.2831	582.78 ± 534.08

3.2 Experimental Setup

- Sliding windows produce supervised pairs $\mathbf{X}_{t-L+1:t} \to \mathbf{X}_{t+1:t+H}$.
- Chronological split: 70% train, 15% validation, 15% test.
- Z-score standardization is fitted only on the training segment: $\tilde{x}_i^{(t)} = (x_i^{(t)} - \mu_i)/\sigma_i$.
- Short–Mid regime: $L \in \{2, 4, 8, 12\}$ and $H \in \{1, 5, 10\}$.
- Long-Range regime: $(L, H) = (50, 20)$; England COVID uses $(20, 10)$.

3.3 Evaluated STGNN Architectures

$$\mathbf{X}' = \mathbf{X}\mathbf{W}_{in}, \quad \mathbf{W}_{in} \in \mathbb{R}^{F\times d}$$

$$\mathbf{U}^{(t,\ell)} = g_\phi\Big(\mathbf{U}^{(t,\ell-1)}, \mathbf{A}(t)\Big), \quad \mathbf{Z}^{(t)} = h_\psi\Big(\{\mathbf{U}^{(\tau,K_s)}\}_{\tau=t-L+1}^{t}\Big)$$

$$\mathbf{h} = \mathrm{Dropout}(\mathrm{ReLU}(\mathbf{Z}^{(t)})), \quad \hat{\mathbf{X}} = \mathbf{h}\mathbf{W}_{out}, \quad \mathbf{W}_{out} \in \mathbb{R}^{d\times H}$$

Model	Family	h_ψ	g_ϕ
CaST	A	Self-att. + FFT	Message passing
GMAN	A	Multi-head attention	Spatial attention
STAEformer	A	Self-attention	Adaptive attention
STGNN	A	Temporal attention	GNN
STGraformer	A	Transformer	Graph attention
TGAT	A	Time encoding + attention	Neighborhood attention
AAGCN	C	Temporal convolution	Adaptive GCN
Graph WaveNet	C	Dilated convolution	Graph convolution
LSGCN	C	GLU + convolution	Chebyshev GCN
MTGNN	C	Dilated/inception conv.	Graph learning
SLCNN	C	Temporal convolution	Chebyshev GCN
STGCN	C	Temporal convolution	Chebyshev GCN
DCRNN	R	GRU	Diffusion convolution
DyGrAE	R	LSTM	GCN
EvolveGCN-H/O	R	GRU	GCN
GCLSTM	R	LSTM	Chebyshev GCN
GConvGRU/LSTM	R	GRU/LSTM	Chebyshev GCN
MPNN-LSTM	R	LSTM	Message passing
TGCN	R	GRU	GCN

3.4 Training Protocol

- AdamW, $B = 32$, up to 200 epochs, early stopping with patience 20.
- 5-epoch warmup followed by cosine annealing; gradient clipping $= 1.0$; mixed precision.
- $S = 10$ independent seeds per configuration; hidden size $d \in \{32, 64\}$.
- Attention heads $= 4$, Chebyshev order $K_{cheb} = 2$, neighborhood size $= 20$, Top-K $= 32$ when applicable.

3.5 Evaluation Protocol

$$\mathrm{MSE} = \frac{1}{NH}\sum_{h=1}^{H}\sum_{i=1}^{N}(x_i^{(h)} - \hat{x}_i^{(h)})^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}},$$

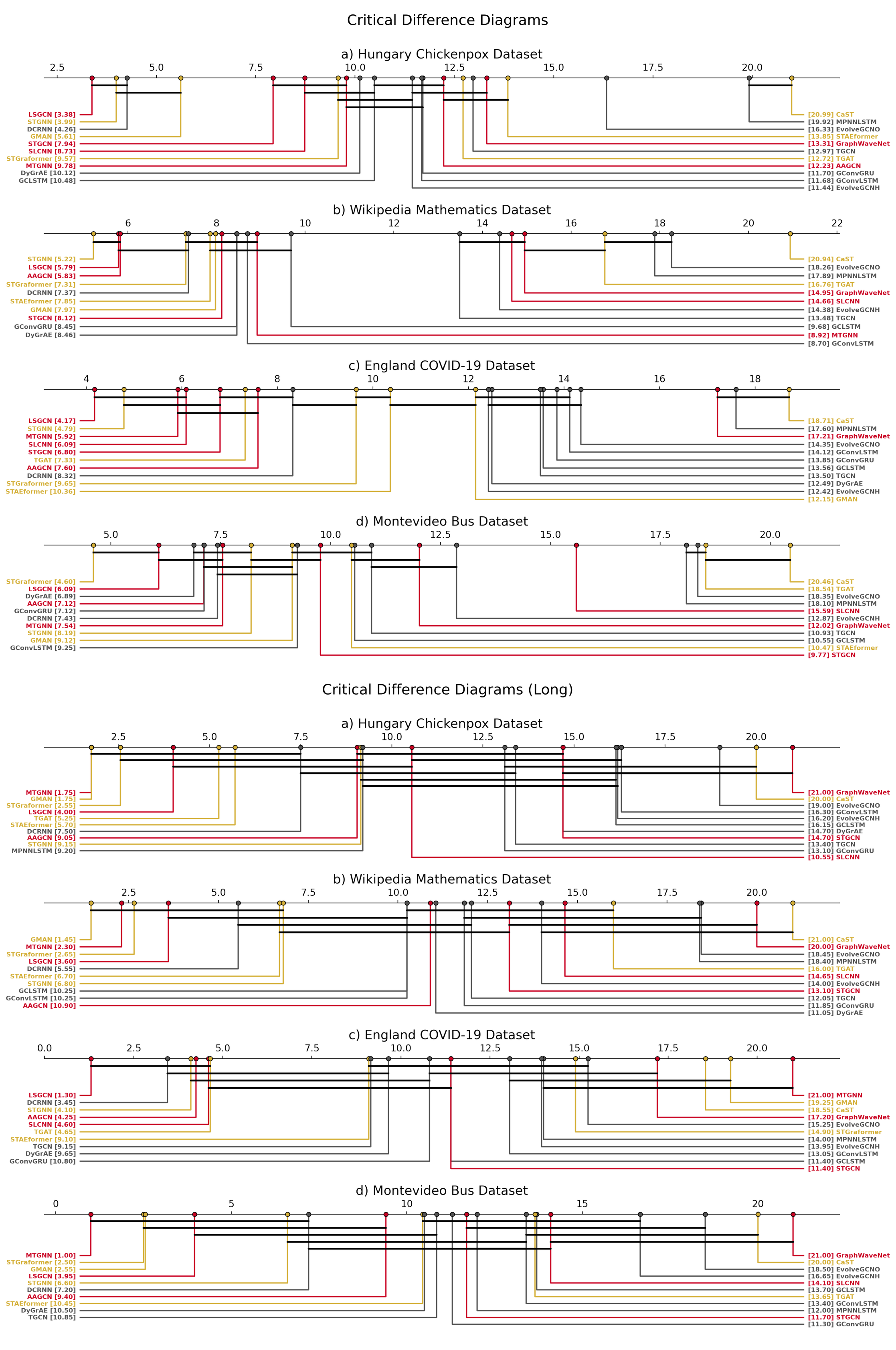
$$\mathrm{MAE} = \frac{1}{NH}\sum_{h=1}^{H}\sum_{i=1}^{N}|x_i^{(h)} - \hat{x}_i^{(h)}|, \qquad R^2 = 1 - \frac{\sum(x-\hat{x})^2}{\sum(x-\bar{x})^2},$$

$$\bar{r}_a = \frac{1}{|S|}\sum_{s\in S} r_{s,a}, \qquad \mathrm{CD} = q_\alpha\sqrt{\frac{A(A+1)}{6|S|}}.$$

$$\Delta\mathrm{RMSE} = \mathrm{RMSE}_{noIP} - \mathrm{RMSE}_{IP}.$$

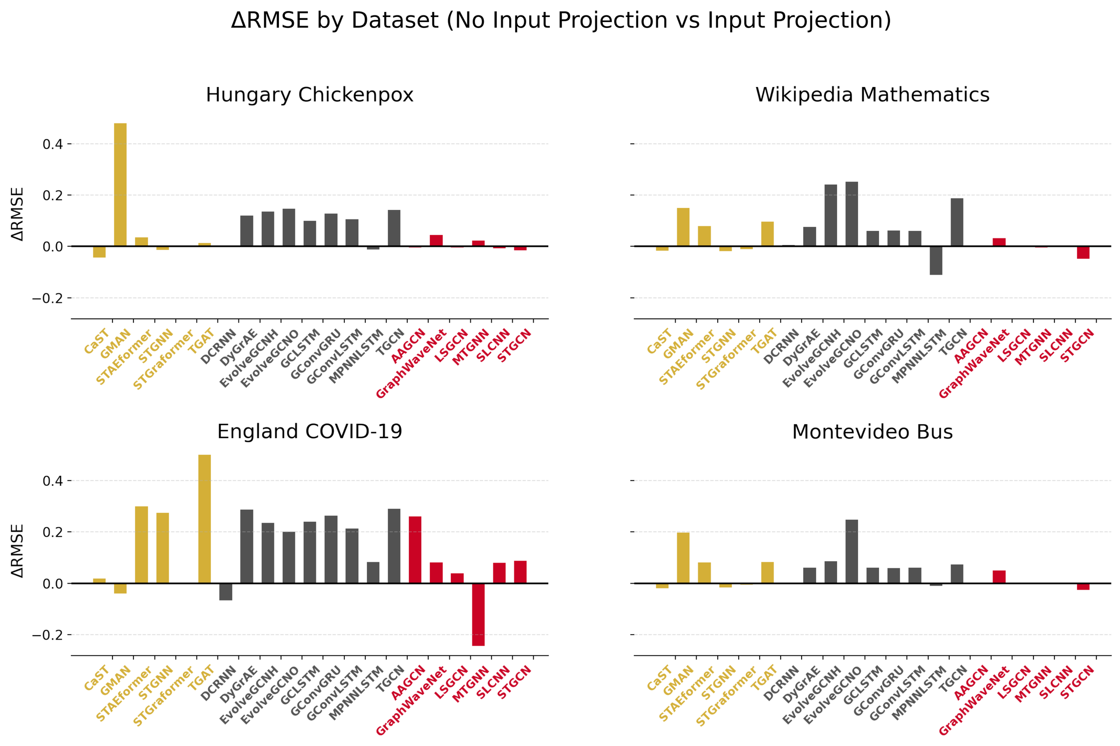
4 Preliminaries Results

4.1 Statistical Ranking



- **Short–Mid:** no universal winner; convolutional models remain competitive.
- **Long-Range:** architectural differences increase; STGraformer/GMAN and MTGNN/LSGCN stand out.
- **Dataset effect:** England COVID and Montevideo show stronger variability due to non-stationarity and topology.

4.2 Input Projection Ablation



- Removing input projection worsens performance in about 63% of cases.
- Stronger degradation appears in recurrent models (86%) than in attention-based (54%) and convolutional models (38%).
- Positive $\Delta\mathrm{RMSE}$ indicates worse performance without the projection layer.

4.3 Spectral–Architectural View

- Low-frequency dominance in WikiMaths and England suggests smoother graph signals, but England remains difficult due to dynamic, directed and non-stationary behavior.
- Montevideo has a balanced spectrum and large diameter, stressing long-range propagation.
- Spectral descriptors contextualize performance but do not imply causal superiority of any family.

5 Conclusions

- A unified benchmark is necessary for fair STGNN comparison across graph domains.
- There is no universally superior architecture: effectiveness depends on dataset, horizon, graph topology, temporal mechanism and input representation.
- Convolutional models are strong and efficient baselines; attention-based models are promising in long-range/non-local regimes; recurrent models are more sensitive to input conditioning.
- Graph spectral analysis provides an interpretable lens for explaining relative model behavior.

Model choice should be dataset-aware: temporal module, spatial operator, graph spectrum and forecasting regime interact.

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